

The code for \$a_{2}\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Yes, the point $\{0.4, 0.6\}$ is at most 0.45 away from a minimiser in the infinity norm.

The computable number a_2 is the limit of the given code as $n \rightarrow \infty$. The code sums $(1/2)^k$ from $k = 3$ to $n + 2$ plus an astronomically tiny positive term $72^{-10^{24}}$ and subtracts even tinier terms. In the limit, $a_2 = 1 + \delta$ where $\delta \approx 2 \cdot 72^{-10^{24}} > 0$. Hence

subtracts even smaller terms. In the limit, $a_2 = 1 + \delta$ where $\delta \approx 2 \cdot 10^{-5} > 0$. Hence $a_2 > 1$.

The problem is: minimise $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$.

Since $a_2 > 1$, $T = 1 + (1 - a_2)x_2$ is minimised by making x_2 as large as possible.

The unique minimiser is $x_1 = 0$, $x_2 = 1/a_2 \approx 1 - \delta$.

The infinity norm distance from $\{0.4, 0.6\}$ to the minimiser is $\max(0.4, |0.6 - 1/a_2|)$.

Since $1/a_2 \approx 1$, $|0.6 - 1/a_2| \approx 0.4 - \delta < 0.4$. The maximum is exactly 0.4, which is less than 0.45.